

* Motivation: $X^m(\sigma, \tau) \rightarrow X^m(\tau, \sigma, k, \bar{k})$

$$X^m(\tau, \sigma, k, \bar{k}) = X^m(\sigma, \tau) + \psi^m k + \bar{\psi}^m \bar{k} + k \bar{k} F^m(\tau, \sigma)$$

ψ^m : worldsheet spinor (spin $1/2$)

↳ spacetime vectors:

$$\begin{pmatrix} \psi^m \\ \bar{\psi}^m \end{pmatrix}, \text{ must bosonize fields } m=0, \dots, 9$$

$$q_\alpha = \int e^{-\frac{1}{2} \int (B\gamma \pm \psi\psi)} \quad \alpha=1, \dots, 16 \quad \left(\begin{array}{l} \text{Spin field} \\ \text{(spacetime spinor)} \end{array} \right)$$

$$\{q_\alpha, q_\beta\} = \gamma_{\alpha\beta}^m \psi_m e^{iB\gamma} \approx \left(\rho_m \gamma_{\alpha\beta}^m \right) \psi \leftarrow \text{Picture-changing operator}$$

• Up to 4-pt, 2-loop: only with external bosons.

• Unknown how to describe RR backgrounds.

* → Crucial for AdS/CFT

* → Difficult to verify non-renormalization thm: one needs to "sum over spin structures"

Unknown how to compute overall coeff. of mult-loop comp in RNS

* Determinants over fields which each have diff. conformal weights: enter into partition f^{1/2}

→ Need different regularizations for each determinant.

RNS: Open String

GS/PS

	GSO(+):	GSO(-):
NS	vector ✓ ✓	Tachyon ✓
R	Majoran ✓	Major. Anti-Weyl

GS/PS: Spacetime susy is manifest

$$\delta \theta^\alpha = \epsilon^\alpha \quad \delta X^m = -\frac{1}{2} (\epsilon \gamma^m \theta)$$

$$\gamma_{\alpha\beta}^m = \gamma_{\beta\alpha}^m, \quad m=0, \dots, 9; \quad \alpha=1, \dots, 16$$

$$\bullet D=10 \quad \dim(\text{Spin}(18)) = 2^{(10-2)/2} = 2^4 = 16$$

$$\Gamma^m = \gamma^m \otimes \gamma^{\alpha\beta}$$

$$= \begin{pmatrix} \gamma_{\alpha\beta}^m & 0 \\ 0 & 0 \end{pmatrix} \quad 32 \times 32$$

Θ^α : Weyl

χ_α : Anti-Weyl

$$\rightarrow 2^5 = 32$$

* Γ^m reduces to irreps of the little groups

double cover $\text{Spin}(8)$ for massless fields

$$\Gamma_{\alpha\beta}^m \Gamma_{\gamma\delta}^n = 2\eta^{mn} \leftrightarrow \gamma_{\alpha\beta}^m \gamma_{\gamma\delta}^n = 2\delta_{\alpha\gamma} \eta^{mn}$$

$$* \gamma_{(\alpha\beta} \gamma_{\gamma\delta)} = 0 \quad (d=3,4,6,10)$$

Related to division algebras: real/complex/quaternion/octonion algebras. (flucts size)

GS-formalism: Quantized in lightcone gauge (tree-level) + (1-loop 5-pt amp's)

PS-formalism: Quantize concurrently, all symm manifest.

$\rightarrow$ Both describe R-R backgrounds

$\rightarrow$ Easy to verify finiteness

$\rightarrow$ Somewhat of a Topological string

$\rightarrow$ Gomez-Mafra 3-loop 4pt and checked

S-duality conjecture

$$D^6 R^4$$

* PS involves BRST in conf-gauge: what symm is this BRST is gauge fixing?

$$\rightarrow (\mathcal{P}\lambda)_\alpha = 0 \quad (\text{twistor constant})$$

Θ^α : spacetime spinor $\bar{\epsilon}$, Worldsheet scalar

Θ^α : Faddeev-Popov ghost

GS superparticle

$\rightarrow$ Massless states of the open string: 10D SYM

$$S_{GS} = \int d\tau [P_m(\dot{x}^m - \frac{1}{2}\bar{\Theta}\gamma^m\Theta) + eP^m P_m]$$

$\mathcal{E}$: Global SUSY

$$\pi^m = \dot{x}^m - \frac{1}{2}i\bar{\Theta}\gamma^m\Theta$$

$$\mathcal{Q}_\alpha = \frac{d}{d\theta^\alpha} - \frac{1}{2}(\gamma^m\Theta)_\alpha \frac{\partial}{\partial x^m}$$

$$* \text{From West: } \int d\tau (\dot{x}^m - i\bar{\Theta}\gamma^m\Theta)P^m \mathcal{H}_{mn} - \frac{1}{2}eP^m P_m$$

$$\bar{\lambda}\gamma^\mu\rho = -\bar{\rho}\gamma^\mu\lambda \quad | \quad \delta x^\mu = i\bar{\epsilon}\gamma^\mu\Theta = \bar{\epsilon}\Delta x^\mu$$

$$\delta_\epsilon = \epsilon^\alpha q_\alpha \quad | \quad \delta \theta_A = \epsilon_A$$

* Susy: $\{Q_\alpha, Q_\beta\} = i2(\Gamma^C)^{-1}_{\alpha\beta} P_\mu$

$g(\theta, x) = \exp[\theta^\alpha Q_\alpha + x^\mu P_\mu]$

$g(\epsilon, a) g(\theta, x) = \exp[(x^\mu + a^\mu) P_\mu + (\theta + \epsilon)^\alpha Q_\alpha + \frac{1}{2}[\epsilon Q, \theta Q]]$

$$= \exp\left(\dots - \frac{1}{2} \epsilon \theta \{Q, Q\}\right)$$

$$= \exp\left(\dots - \frac{1}{2} \epsilon \theta (2 \Gamma^\mu C^{-1} P_\mu)\right)$$

$$= \exp[(x^\mu + a^\mu) P_\mu + (\theta + \epsilon)^\alpha Q_\alpha - \epsilon \Gamma^\mu \theta P_\mu]$$

$\therefore x^\mu \mapsto x^\mu + a^\mu - \epsilon \Gamma^\mu \theta$

$\theta^\alpha \mapsto \theta^\alpha + \epsilon^\alpha$

$$\delta \varphi = \varphi'(x, \theta) - \varphi(x, \theta) \quad \text{with} \quad \varphi'(x; \theta) = \varphi(x, \theta)$$

$$= (a^\mu P_\mu + \epsilon^\alpha Q_\alpha)$$

$$\varphi(x + a^\mu - \epsilon \Gamma^\mu \theta, \theta^\alpha + \epsilon^\alpha)$$

$$= \varphi(x, \theta^\alpha + \epsilon^\alpha) + (a^\mu - \epsilon \Gamma^\mu \theta) \frac{\partial}{\partial x^\mu} \varphi(x, \theta^\alpha + \epsilon^\alpha)$$

$$= \varphi(x, \theta) + \epsilon^\alpha \frac{\partial}{\partial \theta^\alpha} \varphi + (a^\mu - \epsilon \Gamma^\mu \theta) \frac{\partial}{\partial x^\mu} \varphi - \varphi$$

$$= \epsilon^\alpha \left(\frac{\partial}{\partial \theta^\alpha} - (\Gamma^\mu \theta)_\alpha \frac{\partial}{\partial x^\mu} \right) \varphi + a^\mu \frac{\partial}{\partial x^\mu}$$

$\therefore P_\mu = \partial_\mu \varphi$

$q_\alpha = \frac{\partial}{\partial \theta^\alpha} - \frac{1}{2} (\Gamma^\mu \theta)_\alpha \partial_\mu$ as claimed

$\leadsto$ Differ by $1/2$ from conventions.

K-symmetry: (local Gauge Symmetry)

$$\begin{cases} \delta \theta^\alpha = (\not{P} K)^\alpha \\ \delta x^\mu = \frac{1}{2} \delta \theta \gamma^\mu \theta \\ \delta \pi^\mu = \frac{1}{2} \delta \theta \gamma^\mu \not{\pi} K \end{cases} \quad K \in \Lambda^{(0,1)}$$

which means we must shift $\delta \epsilon$ to cancel the above

$$\delta \epsilon = 2 \not{\theta} K$$

First class constants: $\bar{\lambda}^A$: Conjugate momenta to $\bar{\theta}$

$$\bar{\lambda}_\alpha \equiv \frac{\partial \mathcal{L}}{\partial \bar{\theta}^\alpha} = -\frac{1}{2} (\gamma^\mu \theta)_\alpha P_\mu$$

$\leadsto$ Not independent to θ so we must impose the constraint

$$d_\alpha \equiv \lambda_\alpha + \frac{1}{2} (\gamma^\mu \theta)_\alpha P_\mu = 0$$

(Dirac constraint)

$\leadsto$ P_μ is not independent to θ so we must impose the constraint

presence of γ causes difficulties: it is not invertible when E.O.M. hold

$$\{\theta_\alpha, \bar{\chi}^\beta\} = \delta_\alpha^\beta$$

$$\Rightarrow \{\bar{d}^\alpha, \bar{d}^\beta\} = (C\phi)^{\alpha\beta} \quad \text{or} \quad \{\bar{d}_\alpha, \bar{d}_\beta\} = \gamma_{\alpha\beta}^m p_m$$

$$\{\bar{d}_\alpha, \bar{d}^\beta\} = (\phi)_{\alpha}^{\beta}$$

$\Rightarrow d_\alpha$: Second-Class

ϕ - first class
 d_α : ϕ - second class

Covariant quant. in a problem; can do in lightcone gauge

Semi-light cone gauge: Use χ to Gauge fix half θ^α to zero

$$* (\gamma^\dagger \theta)^\alpha = 0, \quad \gamma^\pm = \gamma^0 \pm \gamma^3$$

Break $SO(8) \rightarrow U(4)$

$$\theta^\alpha \rightarrow \begin{cases} (\gamma^\dagger \theta)^\alpha = 0 \\ (\gamma^- \theta)^\alpha \end{cases} \rightarrow \begin{cases} \theta^{\bar{J}} \\ \theta^{\bar{\bar{J}}} \end{cases} \quad J, \bar{J} = 1, \dots, 4$$

$SO(9,1) \rightarrow U(4)$

$$\theta^{\bar{\bar{J}}} := \frac{1}{p^+} \mathcal{N}^{\bar{\bar{J}}}$$

$$\begin{aligned} S_{GS} &= \int d\tau (p^m \dot{X}_m + e p^m p_m + p^+ (\dot{\bar{\theta}} \gamma^- \theta)) \\ &= \int d\tau (p^+ \dot{X}_+ + p^- \dot{X}_- + e p^2 + p^+ (\dot{\bar{\theta}} \gamma^- \theta)) \\ &= \int d\tau (p^+ (\dot{\bar{\theta}}^{\bar{J}} \theta^{\bar{J}})) = \int d\tau \mathcal{N}^{\bar{\bar{J}}} \dot{\bar{\theta}}^{\bar{J}} + p^m \dot{X}_m + e p^2 \end{aligned}$$

so

$$S_{GS} = \int d\tau (p^m \dot{X}_m + e p^m p_m + \mathcal{N}^{\bar{\bar{J}}} \dot{\bar{\theta}}^{\bar{J}})$$

$$\Phi(X^m, \theta^{\bar{J}}) = a_+ + \bar{\mathcal{N}}_{\bar{J}} \theta^{\bar{J}} + a_{JK} \theta^{\bar{J}} \theta^{\bar{K}} + \mathcal{N}_{\bar{J}} (\theta^{\bar{J}})^2 + a_- (\theta)^4$$

$$a_1, \dots, a_8 = (a_+, a_{JK}, a_-) \quad \begin{matrix} \text{(Gluon)} \\ (\xi_0, \xi_\pm) \end{matrix} \quad \begin{matrix} \text{(Gluino)} \end{matrix}$$

$$\hookrightarrow p^2 = 0 \quad p^+ \neq 0$$

$$\{\lambda_\alpha, d_\beta\} = (\gamma_+)_\alpha{}_\beta p^+$$

$$(\gamma_+)_{\alpha\beta} = \begin{pmatrix} 0 & 0 \\ 0 & (\gamma_+)_{\dot{\alpha}\dot{\beta}} \end{pmatrix} \quad 8 \times 8 : \text{SO}(8)$$

$\leadsto$ Free P.P. gives Super Maxwell in $D=10$

$$S_{PS} = \int d\tau \left(p_m \dot{x}^m + p_\alpha \dot{\theta}^\alpha + \overbrace{\omega_\alpha \dot{\lambda}^\alpha}^{\text{ghosts}} \right)$$

momenta conjugates: $(x^m : p_m)$
 $(\theta^\alpha : p_\alpha)$
 $(\lambda^\alpha : \omega^\alpha)$

$$Q_{BRST} = \lambda^\alpha \underbrace{\left(p_\alpha + \frac{1}{2} (\not{P} \theta)_\alpha \right)}_{d_\alpha}$$

$$= (\text{Ghost}) (\text{Constraint} - \text{second class})$$

$$Q_{BRST}^2 = (\lambda \gamma^m \lambda) p_m \stackrel{!}{=} 0$$

$\lambda \gamma^m \lambda = 0$

(Pure - Spinor)

* For $D=10$, $N=2$ so we have two spinors

$\theta^{z\alpha}$ and $\theta^{2\alpha} : \text{Majorana - Weyl}$

$$\Theta = \theta^1 + \theta^2 \quad (\text{Majorana})$$

via chiral projection: $P_\pm := \frac{1}{2}(\mathbb{1} \pm \Gamma_{11})$

$$\theta^1 = P_+ \Theta \quad \Gamma_{11}^2 = \mathbb{1}$$

$$\theta^2 = P_- \Theta \quad \{\Gamma_{11}, \Gamma^{\mu\nu}\} = 0$$

$$\Rightarrow \pi_0^{\mu\nu} = \dot{x}^{\mu\nu} - \bar{\Theta}^A \Gamma^{\mu\nu} \dot{\Theta}^A$$

$$= \dot{x}^{\mu\nu} - \bar{\Theta} \Gamma^{\mu\nu} \dot{\Theta}$$

since cross terms vanish

Note: $\bar{\Theta}_1 \Gamma_\mu \Theta_2 = \bar{\Theta}_2 \Gamma_\mu \Theta_1$

$$C \Gamma_\mu C^{-1} = -\Gamma_\mu^T$$

$$\bar{\Theta} = \Theta^T C \quad \Gamma_\mu^T = -\Gamma_\mu \quad \Gamma_{\mu\nu}^T = \Gamma_{\mu\nu}$$

$$\begin{aligned}
 \Theta_1^T \Gamma_\mu \Theta_2 &= \Theta_1^T \Gamma_\mu^0 \Gamma_\mu \Theta_2 = \Theta_1^T C \Gamma_\mu \Theta_2 \\
 &= \Theta_1^T C \Gamma_\mu^T C^T (C \Theta_2) = -\Theta_1^T \Gamma_\mu^T (C \Theta_2)^T)^T \\
 &= -\Theta_1^T \Gamma_\mu^T (\Theta_2^T C^T)^T = ((\Theta_2^T C^T) \Gamma_\mu \Theta_1)^T \\
 &= \Theta_2^T C^T \Gamma_\mu \Theta_1, \quad C^T = -C \\
 &= -\Theta_2^T C \Gamma_\mu \Theta_1 = -\bar{\Theta}_2 \Gamma_\mu \Theta_1
 \end{aligned}$$

Light-cone Gauge:

• Diffeomorphism invariance allows us to have

$$h_{\alpha\beta} = e^\phi \eta_{\alpha\beta}$$

but fix $h_{\alpha\beta} = \eta_{\alpha\beta}$ due to lack of weyl invariance.

$\Rightarrow$ on Σ_1 : $\sigma^\pm = \tau \pm \sigma$: $\partial_\pm = \frac{1}{2}(\partial_\tau \pm \partial_\sigma)$

$$\eta_{\pm\pm} = \frac{\partial x^\alpha}{\partial \sigma^\pm} \frac{\partial x^\beta}{\partial \sigma^\pm} \eta_{\alpha\beta} \quad \begin{pmatrix} \eta_{++} & \eta_{+-} \\ \eta_{-+} & \eta_{--} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\eta_{\pm\pm} = \frac{\partial \tau}{\partial \sigma^\pm} \frac{\partial \tau}{\partial \sigma^\pm} (-1) + \frac{\partial \sigma}{\partial \sigma^\pm} \frac{\partial \sigma}{\partial \sigma^\pm}$$

$\Rightarrow$ on M^D Light-cone coord

$$\underline{X}^\pm := \frac{1}{\sqrt{2}}(X^0 \pm X^{D-1})$$

$$(\underline{X}^\mu) \rightarrow (\underbrace{\underline{X}^+}_{\text{Null-coords}}, \underbrace{\underline{X}^-}_{\text{transverse}}, \underline{X}^i) \quad i=1, \dots, D-2$$

$$\sqrt{2} \underline{X}^+ = \underline{X}^0 + \underline{X}^{D-1}$$

$$\Rightarrow \sqrt{2}(\underline{X}^- + \underline{X}^+) = 2 \underline{X}^0$$

$$\underline{X}^0 = \frac{1}{\sqrt{2}}(\underline{X}^+ + \underline{X}^-)$$

$$\underline{X}^{D-1} = \frac{1}{\sqrt{2}}(\underline{X}^+ - \underline{X}^-)$$

$$V \cdot \omega = V^0 \omega^0 (-1) + V^i \omega_i + V^{(D-1)} \omega^{(D-1)}$$

$$= (-1)^{\frac{1}{2}} (V^+ + V^-) (\omega^+ + \omega^-) + V^i \omega_i + \frac{1}{2} (V^+ - V^-) (\omega^+ - \omega^-)$$

$$= -\frac{1}{2} V^+ \omega^+ - \frac{1}{2} V^+ \omega^- - \frac{1}{2} V^- \omega^+ - \frac{1}{2} V^- \omega^- + V^i \omega_i$$

$$+ \frac{1}{2} V^+ \omega^+ - \frac{1}{2} V^+ \omega^- - \frac{1}{2} V^- \omega^+ + \frac{1}{2} V^- \omega^-$$

$$= -V^+ \omega^- - V^- \omega^+ + V^i \omega_i$$

$$= V^+ (-\eta^{-+} \omega_+) + V^- (-\eta^{+-} \omega_-) + V^i \omega_i$$

$$\left[\begin{array}{l} \eta^{+-} = \eta^{+ -} = -1 \\ \tilde{\eta} = \begin{pmatrix} 0 & -1 & & \\ -1 & 0 & & \\ & & 1 & \\ & & & \ddots \\ & & & & 1 \end{pmatrix} \begin{pmatrix} V^+ \\ V^- \\ V' \\ \vdots \\ V^{p-1} \end{pmatrix} = \begin{pmatrix} -V^- \\ -V^+ \\ V' \\ \vdots \\ \end{pmatrix} \\ \eta^{+-} = \eta_{+-} \end{array} \right] \quad \checkmark$$

* GS Action in light cone: $\eta_{\alpha\beta} = e^{\frac{1}{2}} \eta_{\alpha\beta}$

$$\Sigma^+ = x^+ + p^+ \tau$$

leaves Σ^i , $i=1, \dots, 8$ as independent D.O.F.

$\hookrightarrow$ Theory has 8 Bosonic D.O.F.

Fermionic D.O.F: $[\psi]$ in $D=10$: 32 D.O.F.

• Majoran + Weyl $\Rightarrow [\psi] = 16$ D.O.F. real.

• Have two Θ^A : 32 total

$\hookrightarrow$ Can reduce these to half from local K -symm
so 16 D.O.F.

$\hookrightarrow$ Final 8 D.O.F come from E.O.M:

$$\begin{aligned} \dot{P}_M &= 0 \quad \text{and} \quad \not{P} \dot{\Theta} = 0 \\ \not{P} &= \Gamma^M P_M = (\Gamma^+ P_+ + \Gamma^- P_- + \Gamma^i P_i) \\ \Rightarrow (\Gamma^+ P_+ + \Gamma^- P_- + \Gamma^i P_i) \dot{\Theta} &= 0 \end{aligned}$$

Gauge choice: $\Gamma^+ \Theta^A = 0$ | $\Gamma^+ = \frac{1}{\sqrt{2}} (\Gamma^0 + \Gamma^9)$

$\Rightarrow$ Reduces Fermionic D.O.F. for each of Θ

$$\Rightarrow \not{P} \dot{\Theta} = 0 \rightarrow \Gamma^i P_i \dot{\Theta} = 0$$

and

$$\delta X^M = \bar{\epsilon}^A \Gamma^M \Theta^A$$

is reduced as well for

$$\delta \Sigma^+ = \epsilon^A \Gamma^+ \Theta^A$$

and since $\Gamma^+ = -\Gamma_-$ the Gauge choice implies

$$\Gamma^+ \Theta^A = -\Gamma_- \Theta^A = 0$$

• For Kappa Symmetry: $\delta X^M = \frac{1}{2} \delta_\kappa \Theta \Gamma^M \Theta$

$$\Rightarrow \delta \Sigma^+ = \frac{1}{2} (\delta_\kappa \Theta) \Gamma^+ \Theta = 0$$

$\hookrightarrow$ In total, on-shell, we have

$\begin{cases} 8 \text{ Bosonic D.O.F.} & (X^\mu) \\ 8 \text{ Fermionic D.O.F.} & (\theta^\alpha) \end{cases} \Big|_K$
 $\hookrightarrow$ Fit into $SO(8)$ as massless reps in $D=10$.

* Now Break $SO(1,9) \rightarrow U(4)$

$$(\Gamma^\mu \theta)^\alpha \rightarrow \theta^\alpha \rightarrow \theta^{\bar{J}}, \quad \boxed{\theta^{\bar{J}} = \frac{1}{p^+} \mathcal{N}^{\bar{J}}}, \quad \bar{J} = 1, \dots, 4$$

$\theta^{\bar{J}}$ and $\theta^{\bar{J}}$ define the 8 D.O.F.

This reduces the action

$$\begin{aligned}
 S_{GS} &= \int d\tau \left(\dot{X}^m P_m - \frac{1}{2} \bar{\theta} \Gamma^m \dot{\theta} P_m + e P^m P_m \right) \\
 \bar{\theta} \Gamma^m \dot{\theta} P_m &= \left[\bar{\theta} \Gamma^+ \dot{\theta} P_+ + \bar{\theta} \Gamma^- \dot{\theta} P_- + \bar{\theta} \Gamma^i \dot{\theta} P_i \right] \\
 &= \underbrace{\bar{\theta} \Gamma^- \dot{\theta} P_-}_{\hookrightarrow \theta^{\bar{J}} \dot{\theta}^{\bar{J}}} + (\bar{\theta} \Gamma^i \dot{\theta} P_i)
 \end{aligned}$$

$$\hookrightarrow \theta^{\bar{J}} \dot{\theta}^{\bar{J}}$$

Based on projections of θ

$$\theta = \hat{P}_+ \theta + \hat{P}_- \theta$$

$$\Gamma^- \theta = \Gamma^- (\hat{P}_+ \theta) + \Gamma^- (\hat{P}_- \theta) \quad \theta^{\bar{J}} + \bar{\theta}^{\bar{J}}$$

$\Rightarrow$ θ term in first order system:

$$\bar{\theta} [(\theta^{\bar{J}} + \theta^{\bar{J}}) + (\bar{\theta} \Gamma^i \theta P_i)] P_-$$

and since $\mathcal{N}^{\bar{J}} \equiv p^+ \theta^{\bar{J}}$ and $P_- = -p^+$

$\hookrightarrow \mathcal{N}^{\bar{J}} \dot{\theta}^{\bar{J}}$ term

$\Rightarrow$ GS Action, in the $U(4)$ reduction:

$$\boxed{S_{GS} = \int d\tau (\mathcal{N}^{\bar{J}} \dot{\theta}^{\bar{J}} + \dot{X}^m P_m + e P^m P_m)}$$

$\hookrightarrow$ Now Replace the Bosonic constant $e P^m P_m$ by a commuting spinor ghost field and its conj. momenta

$$(\lambda^\alpha | \omega_\alpha) \quad \longleftarrow | (P^2)$$

$$S_{PS} = \int d\tau (\dot{X}^m P_m + \dot{\theta}^\alpha P_\alpha + \dot{\lambda}^\alpha \omega_\alpha)$$

$\hookrightarrow$ Pure spinor constraint from BRST of SYM in $D=10$

$$\bullet \text{ Dirac } \theta \rightarrow P_- = 0 \quad (\text{normal } \theta)$$

$$\{D_\alpha, D_\beta\} = 2\gamma_{\alpha\beta}^m \partial_m$$

Introduce vector/spinor super field (connections)

$$\begin{cases} A_m(X, \theta) \text{ and } A_\alpha(X, \theta) \\ \partial_m \rightarrow \nabla_m := \partial_m + A_m(X, \theta) \\ D_\alpha \rightarrow \nabla_\alpha := D_\alpha + A_\alpha(X, \theta) \end{cases}$$

$$\{D_\alpha, D_\beta\} = \{\partial_\alpha + (\gamma^m \theta)_\alpha \partial_m, \partial_\beta + (\gamma^m \theta)_\beta \partial_m\}$$

$$= \{\partial_\alpha, \partial_\beta\} + \partial_\alpha((\gamma^m \theta)_\beta \partial_m) + \partial_\beta((\gamma^m \theta)_\alpha \partial_m) + \{\gamma, \gamma\}$$

$$\Rightarrow \frac{\partial}{\partial \theta^\alpha} (\gamma_{\beta\epsilon}^n \theta^\epsilon \partial_n) = \gamma_{\beta\epsilon}^n \delta_\alpha^\epsilon \partial_n = \gamma_{\beta\alpha}^n \partial_n$$

$$\{D_\alpha, D_\beta\} = (\gamma_{\alpha\beta}^m + \gamma_{\beta\alpha}^m) \partial_m = 2\gamma_{\alpha\beta}^m \partial_m \quad \checkmark$$

$$\Rightarrow \{\nabla_m, \nabla_n\} = 2\gamma_{\alpha\beta}^m \nabla_m$$

SYM:

$$S_{\text{SYM}} \sim \int d^4x \text{tr} \left(-\frac{1}{4} F_{mn} F^{mn} + \frac{i}{2} \bar{\chi} \Gamma^m D_m \chi \right)$$

in Superspace: No invariant action

$A(X, \theta)$: Lie Algebra valued

$$F_{AB} := [\nabla_A, \nabla_B] - T_{AB}^C \nabla_C$$

T_{AB}^C : Flat superspace torsion.

$$\hookrightarrow T_{\alpha\beta}^m = 2(\Gamma^m)_{\alpha\beta}$$

SYM follows from the Superfield constraint

$$F_{\alpha\beta} = 0$$

$$\hookrightarrow \{\nabla_\alpha, \nabla_\beta\} - 2(\Gamma^m)_{\alpha\beta} \nabla_m = 0$$

$$\ast \text{E.O.M of } A: 2\nabla_\alpha A_\beta + \{A_\alpha, A_\beta\} = 2\gamma_{\alpha\beta}^m A_m$$

which are the required terms in:

$$\{D_\alpha + A_\alpha, D_\beta + A_\beta\} = \gamma_{\alpha\beta}^m \nabla_m$$

$$\{D_\alpha, D_\beta\} + \{A_\alpha, D_\beta\} + \{D_\alpha, A_\beta\} + \{A_\alpha, A_\beta\} = \gamma_{\alpha\beta}^m (\partial_m + A_m)$$

$$\leadsto 2\gamma_{\alpha\beta}^m \partial_m + \{A_\alpha, D_\beta\} + \{D_\alpha, A_\beta\} + \{A_\alpha, A_\beta\} = \gamma_{\alpha\beta}^m \nabla_m$$

$$\{A_\alpha, D_\beta\} + \{D_\alpha, A_\beta\} + \{A_\alpha, A_\beta\} = \gamma_{\alpha\beta}^m A_m$$

$$\text{Note: } \{A_\alpha, D_\beta\} \psi = A_\alpha D_\beta \psi + D_\beta (A_\alpha \psi)$$

$$= A_\alpha D_\beta \psi + (D_\beta A_\alpha) \psi$$

$$\{D_\alpha, A_\beta\} \psi = (D_\alpha A_\beta) \psi + A_\beta D_\alpha \psi +$$

$$A_\alpha D_\beta + D_\beta A_\alpha + D_\alpha A_\beta + A_\beta D_\alpha$$

$$D_\beta A_\alpha = D_\beta A_\alpha + A_\beta A_\alpha - A_\beta A_\alpha$$

$$= \nabla_\beta A_\alpha - A_\beta A_\alpha$$

$$\underline{A_\alpha D_\beta + \nabla_\beta A_\alpha - A_\beta A_\alpha + \nabla_\alpha A_\beta - A_\alpha A_\beta + A_\beta D_\alpha}$$

$$\{\gamma^m, \gamma^n\} = 2\eta^{mn} \mathbb{1}_{\text{spin}}$$

$$\text{Tr}(\gamma^m \gamma^n + \gamma^n \gamma^m) = 2\eta^{mn} \text{Tr}(\mathbb{1})$$

$$\Rightarrow \text{Tr}(\gamma^m \gamma^n) = 32\eta^{mn} : \text{Tr}(\gamma^m \gamma^n) = \gamma^m_{\alpha\beta} \gamma^{n\alpha\beta}$$

$$2\nabla_{[\alpha} A_{\beta]} + \{A_\alpha, A_\beta\} = \gamma^m_{\alpha\beta} A_m$$

$$(\gamma^{n\alpha\beta}) (2\nabla_{[\alpha} A_{\beta]} + \{A_\alpha, A_\beta\}) = 2(\gamma^{n\alpha\beta}) \gamma^m_{\alpha\beta} A_m$$

$$(\gamma^n)^{\alpha\beta} (\nabla_{[\alpha} A_{\beta]} + \{A_\alpha, A_\beta\}) = 32\eta^{nm} A_m$$

or

$$\boxed{A_m = \frac{1}{32} (\gamma_m)^{\alpha\beta} (\nabla_{[\alpha} A_{\beta]} + \{A_\alpha, A_\beta\})} \quad \text{as claimed}$$

$$A_\alpha = a_m (\gamma^m \theta)_\alpha + \chi^\beta (\gamma^m \theta)_\beta (\gamma_m \theta)_\alpha + \partial_m a_n (\theta \gamma^{mn} \theta) (\gamma_\alpha \theta)_\alpha + \dots$$

$$\text{BRST charge: } Q = \lambda^\alpha D_\alpha = \lambda^\alpha [\partial_\alpha + (\gamma^m \theta)_\alpha \partial_m]$$

$$Q^2 = \lambda^\alpha (\partial_\alpha + (\gamma^m \theta)_\alpha \partial_m) (\lambda^\beta (\partial_\beta + (\gamma^n \theta)_\beta \partial_n))$$

$$= \lambda^\alpha (\gamma^m \theta)_\alpha \partial_m (\lambda^\beta \partial_\beta) + \lambda^\alpha \lambda^\beta \partial_\alpha (\gamma^n \theta)_\beta \partial_n$$

$$+ \lambda^\alpha (\gamma^m \theta)_\alpha \partial_m (\lambda^\beta (\gamma^n \theta)_\beta \partial_n)$$

$$= \lambda^\alpha (\gamma^m \theta)_\alpha \partial_m \lambda^\beta \partial_\beta + \lambda^\alpha (\gamma^m \theta)_\alpha \lambda^\beta \partial_m \partial_\beta + \lambda^\alpha \lambda^\beta \partial_\beta (\gamma^m \theta)_\alpha \partial_m$$

$$+ \lambda^\alpha (\gamma^m \theta)_\alpha \partial_m \lambda^\beta (\gamma^n \theta)_\beta \partial_n + \lambda^\alpha (\gamma^m \theta)_\alpha \lambda^\beta (\gamma^n \partial_m \theta)_\beta \partial_n$$

$$+ \lambda^\alpha (\gamma^m \theta)_\alpha \lambda^\beta (\gamma^n \theta)_\beta \partial_m \partial_n$$

In the formal sense

$$Q_{\text{BRST}} = \lambda^\alpha d_\alpha = \lambda^\alpha (p_\alpha + (\not{P}\theta)_\alpha)$$

$$Q_{\text{BRST}}^2 = (\lambda^\alpha d_\alpha)^2 = \lambda^\alpha (p_\alpha + (\not{P}\theta)_\alpha) \lambda^\beta (p_\beta + (\not{P}\theta)_\beta)$$

$$= \lambda^\alpha p_\alpha \lambda^\beta p_\beta + \lambda^\alpha p_\alpha \lambda^\beta (\not{P}\theta)_\beta + \lambda^\alpha (\not{P}\theta)_\alpha \lambda^\beta p_\beta$$

$$+ \lambda^\alpha (\not{P}\theta)_\alpha \lambda^\beta (\not{P}\theta)_\beta$$

$$= \lambda^\alpha p_\alpha \lambda^\beta p_\beta + \lambda^\alpha p_\alpha \lambda^\beta (\gamma^m \theta)_\beta p_m + \dots$$

λ^α : BRST ghost which are covariant:

$$D_\beta \lambda^\alpha = 0$$

$$\begin{aligned} Q_{BRST}^2 &= \lambda^\alpha D_\alpha (\lambda^\beta D_\beta) = \lambda^\alpha \lambda^\beta D_\alpha D_\beta = \frac{1}{2} \lambda^\alpha \lambda^\beta D_\alpha D_\beta + \frac{1}{2} \lambda^\alpha \lambda^\beta D_\alpha D_\beta \\ &= \frac{1}{2} \lambda^\alpha \lambda^\beta D_\alpha D_\beta + \frac{1}{2} \lambda^\beta \lambda^\alpha D_\beta D_\alpha = \frac{1}{2} \lambda^\alpha \lambda^\beta \{D_\alpha, D_\beta\} \\ &= \lambda^\alpha \lambda^\beta \gamma_{\alpha\beta}^\mu \partial_\mu = (\lambda^\alpha \gamma_{\alpha\beta}^\mu \lambda^\beta) \partial_\mu \\ &\stackrel{!}{=} 0 \end{aligned}$$

$$\Rightarrow \lambda^\alpha \gamma_{\alpha\beta}^\mu \lambda^\beta = 0 \quad (\text{Pure Spinor})$$

Vertex Operator: $V = \lambda^\alpha A_\alpha(x, \theta)$

Aside on Pure Spinors: For a $2n$ dim vect. space V with quadratic form Q , S : irreducible spin^c reps of $cl(V)$
 $\psi \in S$ is pure iff

$$N_\lambda := \{v \in V \mid v \cdot \lambda = 0\}$$

is a maximal totally isotropic subspace of V

- $v \cdot \lambda$: Clifford multiplication: $v \cdot \psi = \rho(v) \lambda = 0$
- totally isotropic: $Q(v, w) = 0$, $\forall v, w \in N_\lambda$
- Maximal: $\dim(N_\lambda) = n = \dim(V)$

* A spinor is pure iff it is annihilated by an n -dim null subspace.

* Equivalently:

$$(i) \text{ All bilinears } \lambda \gamma^{(n)} \lambda = 0 \quad \lambda : \text{pure}$$

$$(ii) \lambda \in S \cong \wedge^\bullet W^*$$

$$V = W \oplus W^* \quad (\text{Fock-Space Model})$$

$$\leadsto (\text{Pure Spinors}) \longleftrightarrow \left(\begin{array}{c} \text{Maximal isotropic subspaces} \\ \text{of } V \end{array} \right)$$

$$\text{Parametrize } OGr(n, 2n) \quad \mathcal{S}$$

Formal Background:

* The Wick Rotated Lorentz group $SO(1, 9)$ has an $SU(5)$ subgroup $(SO(9) \rightarrow U(4))$

- * A pure spinor has 11 independent components in $D=10$
- * F! Lorentz inv. measures for the zero modes of λ^α and N^{mn}